A Review of Student-led Initiatives in European Space Activities

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Motivation

University-guided projects

CSUT

CSUM

UNIVERSITAT POLITÈCNICA DE CATALUNYA BARCELONATECH

NANOSAT LAB

Student self-organization

UKSEDS

UK STUDENTS FOR THE EXPLORATION AND DEVELOPMENT OF SPACE

Bundesverband studentischer Raumfahrt

SEDS Italy

Student Competitions of Interest

- Four domains
  - Microgravity
  - Satellites
  - Rockets
  - Rovers
- European
  - In contrast: National competitions

REXUS/BEXUS
Student Experiment Programme

FLY YOUR SATELLITE!

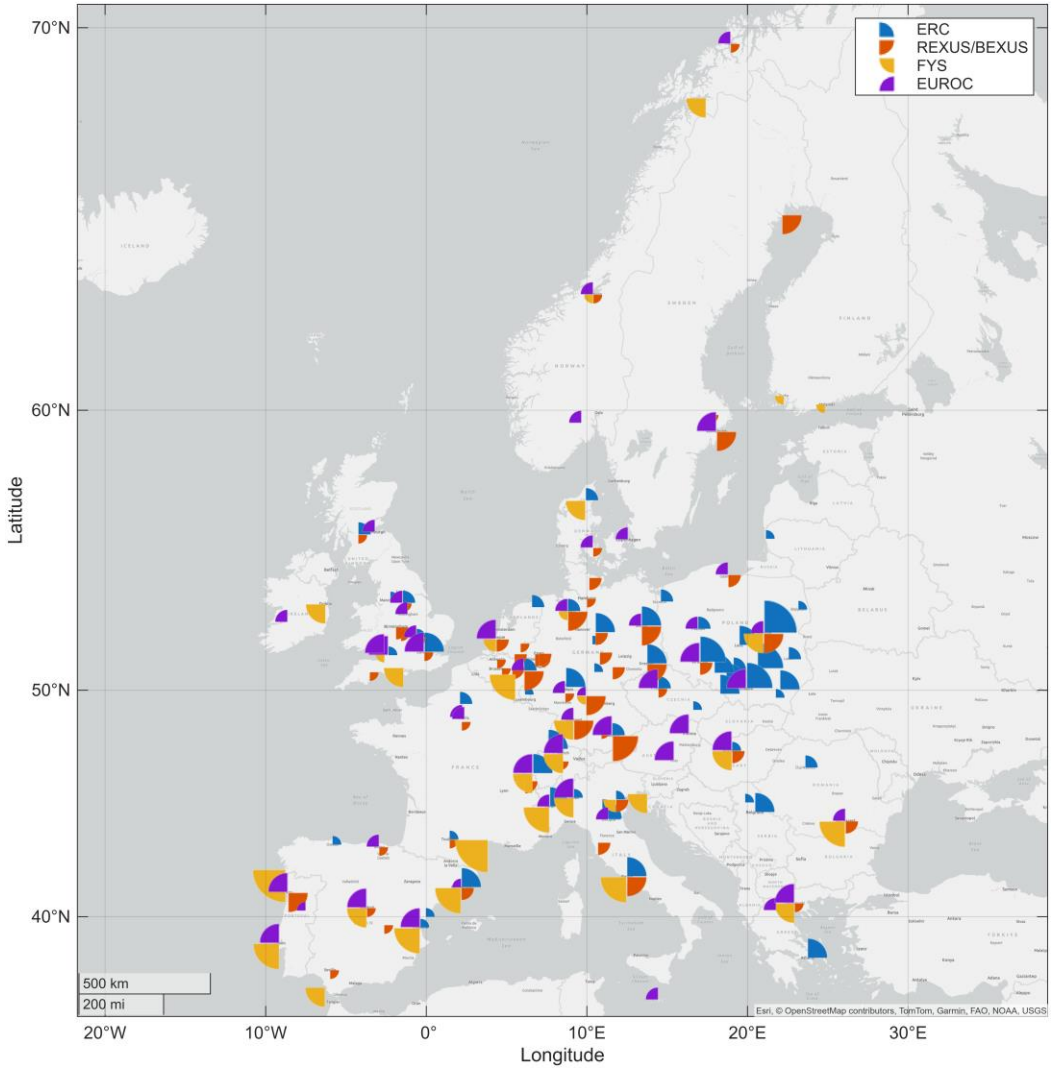
EUROC
EUROPEAN ROCKETRY CHALLENGE

ERC
EUROPEAN ROVER CHALLENGE

Distribution of projects

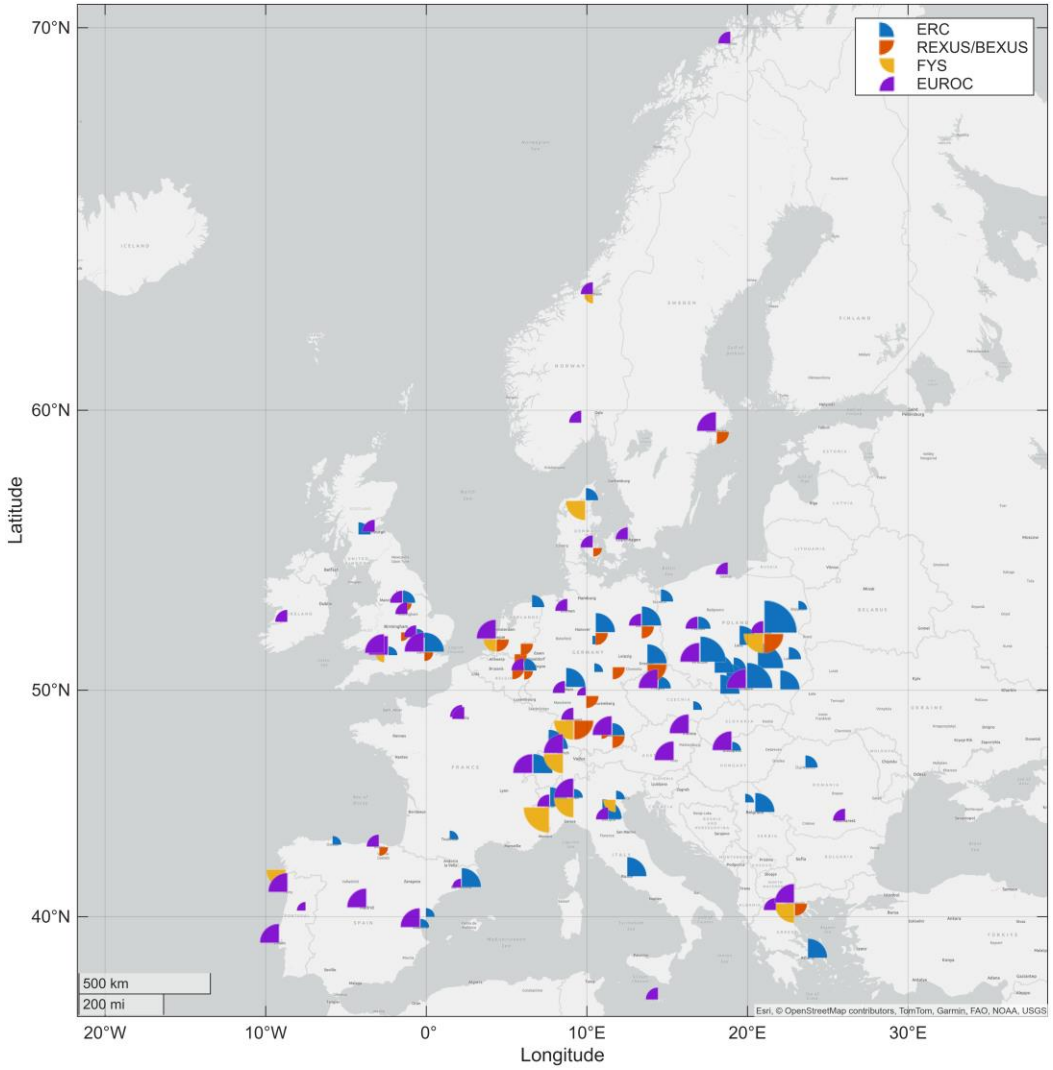
All of Europe is present (almost) – but not equally

Many smaller universities too!



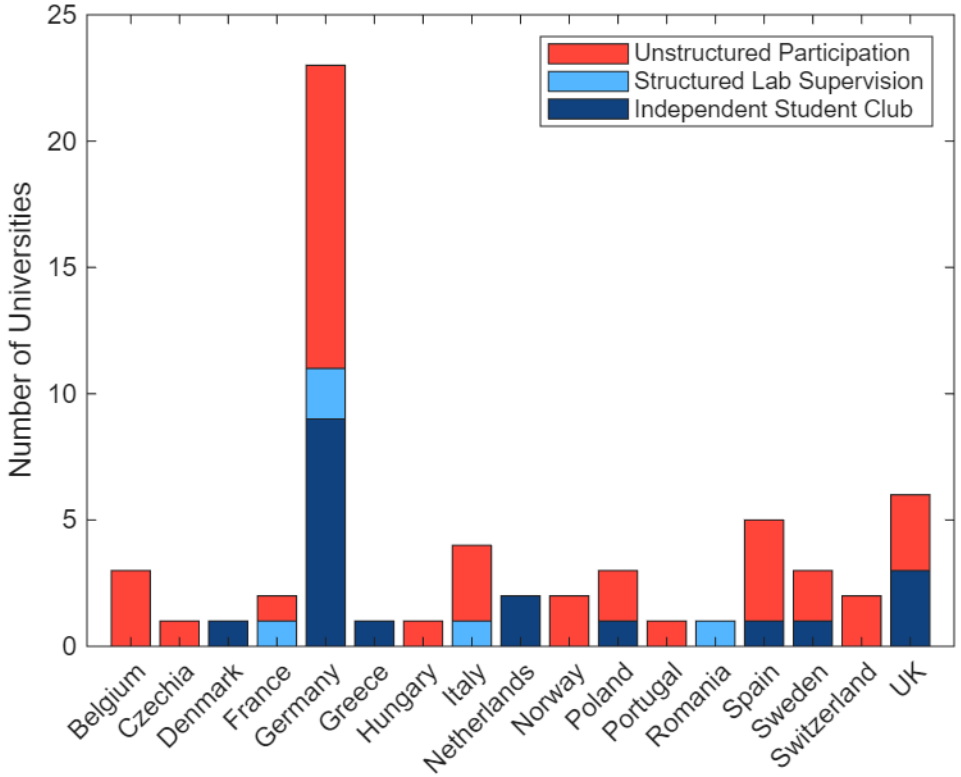
Distribution of projects - Student teams only -

But many projects have no student association backing them



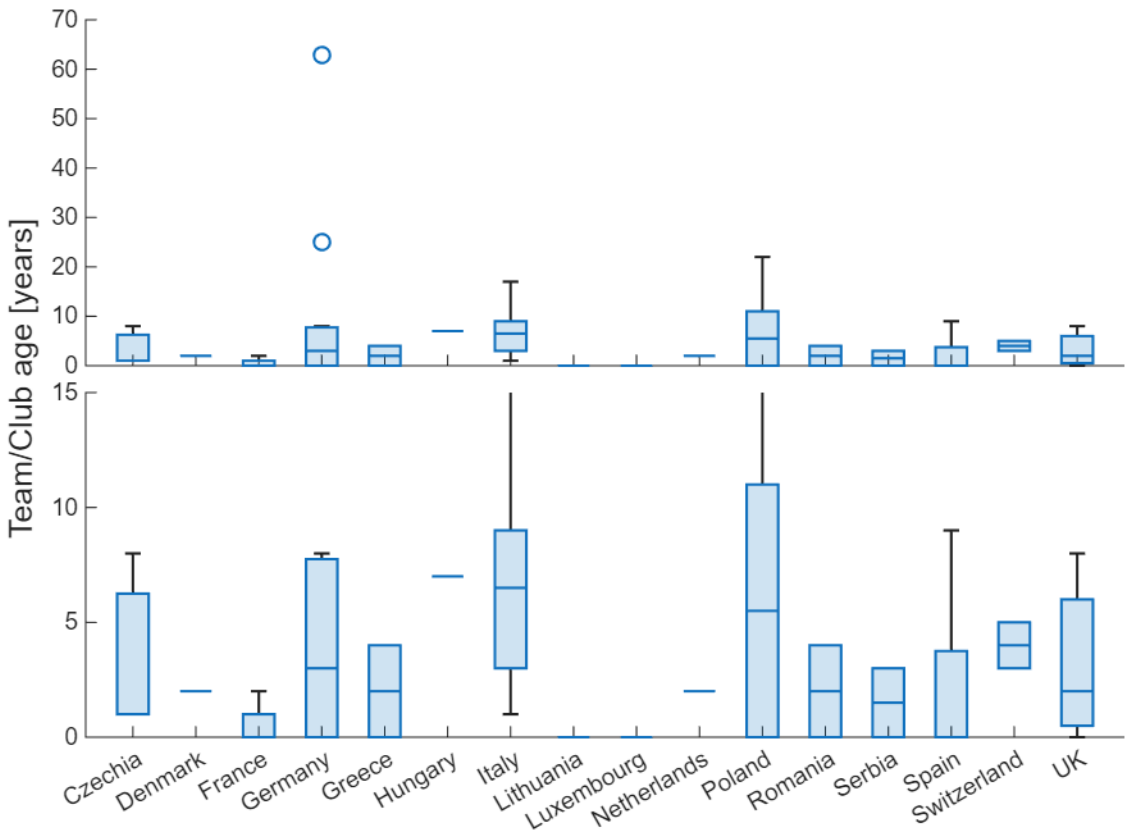
REXUS/BEXUS: Different Team Structures

- Independent student teams prevalent in Germany, Netherlands, UK
- University-provided structures also found
- Most teams dissolve after project is finished!
- Others create proper student associations → **Persistence**



ERC: Participation and Team Lifetime

- Almost all ERC teams are student-led and -organized
- Some teams are very old (but not necessarily in the field of robotics)
- New teams still arrive!



EUROPEAN ROVER CHALLENGE

FYS and EUROC: Polar opposites

FYS

- Largely faculty-backed – even more than REXUS/BEXUS
- Multi-year projects
- Long cycles

➢ Barrier of entry: **High**

EUROC

- Independent student initiatives common – like ERC
- Persistent participation
- High rate of iterations

➢ Barrier of entry: **Low**

Conclusion

- Independent student associations enable sustained hands-on space educational acitivities
- Alternative: Structured faculty support
  - BUT: Less student autonomy!

- Rocketry and robotics: Most accessible
- Fly Your Satellite & REXUS/BEXUS: Good professional guidance

Questions?

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Maps are available as zoomable Matlab files at:

https://github.com/isae-vonarnim/ssea_student_teams_repository
(See paper for clickable link)